

United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4889: Machine Learning

Final Exam: Spring 2025 Time: 2 Hours Marks: 40

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Answer all of the following questions.

1. a) How do you deal with imbalanced datasets where certain classes are underrepresented? Perform the stratified k-fold cross-validation on the following dataset, where $k=3$. [5]

ID	Temperature	Hungry	Air quality	Play
1.	Low	No	Fair	No
2.	Low	Yes	Excellent	No
3.	Medium	Yes	Fair	Yes
4.	High	No	Fair	Yes
5.	High	Yes	Fair	Yes
6.	High	Yes	Fair	No
7.	Low	Yes	Excellent	Yes
8.	Medium	No	Excellent	No
9.	Medium	Yes	Fair	Yes
10.	High	No	Excellent	No

- b) Fit the Random Forest Regression model to predict $\mathbf{B}$ based on $\mathbf{A}$ where each decision tree (T) is trained on minimum four different subsets generated from the following dataset. Moreover, the final prediction is defined as follows: [5]

$$\hat{B} = \frac{1}{T} \sum_{i=1}^T f_i(A)$$

$\mathbf{A}$	3	5	7	11	15	23
$\mathbf{B}$	8	12	16	20	24	32

2. a) Classify the data into three classes using competitive learning. The training data samples are given in the following table. [10]

Training samples

i1	0	1	0
i2	0	0	1
i3	1	1	0

Weight matrix is given as follows:

$$w = \begin{pmatrix} 0.2 & 0.6 & 0.5 \\ 0.8 & 0.4 & 0.7 \\ 0.5 & 0.5 & 0.6 \end{pmatrix}$$

Test the unseen data sample $i_4 = (1, 0, 1)$.

3. a) Draw the hidden states (s) structure from the following transition probabilities: [10]

$$T_{ij} = \begin{pmatrix} 0.7 & 0.3 & 0 \\ 0 & 0.7 & 0.3 \\ 0 & 0 & 1 \end{pmatrix}$$

Observation number of island in the vertical slice for image **K**:

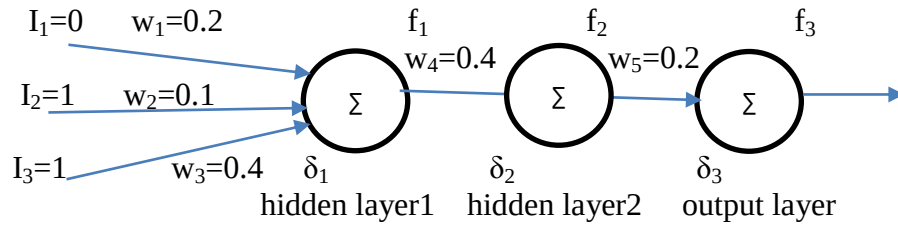
$$K_{ij} = \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.8 & 0.1 & 0.1 \\ 0.3 & 0.4 & 0.3 \end{pmatrix}$$

Observation number of island in the vertical slice for image **Q**:

$$Q_{ij} = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0 & 0.2 & 0.8 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$$

Let us consider an image is segmented into 4 slices as {3, 1, 2, and 1}. What image is more likely to produce this observation sequence, image **K** or **Q**?

4. a) The multilayer neural network is given below with 5 weights: [10]



Initial weight changes are as follows: $\Delta w_1=0.2$, $\Delta w_2=-0.1$, $\Delta w_3=0.5$, $\Delta w_4=0.3$, $\Delta w_5=-0.2$. Input training data pattern is defined as $I_1=0$, $I_2=1$, $I_3=1$, $d=1$. Determine the output Y_k for the test data pattern: $I_1=1$, $I_2=0$, $I_3=1$ where momentum co-efficient and learning rate are $\mu=0.35$ and $\eta=0.001$, respectively.